# Vipul Sapra

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## EDUCATION

**University of Cincinnati, Carl H. Lindner College of Business** — Cincinnati, OH
Master of Science in Information Systems and Graduate Certificate in Data Science — GPA: 3.95/4
April 2022

**Pune University, Maharashtra Institute of Technology** — Pune, India
Bachelor of Engineering in Petrochemical — June 2017

## TECHNICAL SKILLS

**Concepts:**Generative AI & LLMs, Agentic AI Architectures, Machine Learning & Predictive Modeling, MLOps, Autonomous Anomaly Detection, Time-Series Forecasting (SARIMA), Retrieval-Augmented Generation (RAG), Conversational AI Evaluation, Dimensional Data Architecture (Star/Snowflake), Enterprise Data Governance & Lineage.

**Programming Languages and Tools:**Agentic AI (KIRO IDE, GitHub Copilot, Prompt Engineering), Python (PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas, NumPy), SQL (Advanced Query Optimization, Stored Procedures), Databricks, Snowflake, AWS (Redshift, S3), Azure (ADF, Synapse), Apache Spark / PySpark, Power BI (DAX), Tableau, MicroStrategy, Airflow, Collibra, One Automation, JIRA, Git/GitHub, Microsoft Excel.

**Certifications:** Data Science Graduate Certificate â€“ University of Cincinnati

## PROFESSIONAL EXPERIENCE

**Charter Communications** — Charlotte, NC
**Data Scientist & AI Engineer** — May 2022
â€“ Present

- **Engineered an enterprise AI-driven reporting platform** that autonomously answers ad-hoc data questions, streamlining scheduled report delivery and reducing manual reporting turnaround for business stakeholders. **Agentic AI powered KIRO IDE**.
- **Engineered a custom Data Profiler tool** using **Agentic AI powered KIRO IDE** to automate the initial assessment of raw datasets, reducing the time spent on manual data discovery by over 40%.
- **Built and deployed XGBoost classification models** to predict customer support-seeking behaviors and repair calls, contributing to a **20% improvement in service reliability.**
- **Engineered and deployed an autonomous AI anomaly detection framework**, identifying schema variations and data drift with **95% precision** to safeguard downstream enterprise BI pipeline integrity.
- **Architected advanced predictive models (SARIMA)** to forecast truck rolls for customers, optimizing workforce capacity planning and **reducing response times by 20%**.

**ETL Developer**

- **Architected a modular Data Quality Audit framework** that enables the seamless integration of new audit logic, creating a scalable and maintainable suite that serves as the organizationâ€™s primary data integrity engine.
- **Engineered comprehensive Data Quality dashboards** to facilitate a seamless migration of BI assets from Teradata to AWS Redshift, maintaining 100% data integrity during the transition.
- **Engineered an automated, notification-driven Data Quality Audit framework** that streamlined short-term validation cycles; reduced operational turnaround time by **over 80%** while ensuring real-time data reliability.

**Business Intelligence Analyst**

- **Designed Key Operating Metric (KOM) Dashboards** for executive leadership, synthesizing complex datasets into high-level visualizations to drive data-informed corporate strategy.
- **Engineered multi-layered Data Quality Audits** leveraging **Stored Procedures, MicroStrategy, and Tableau**; implemented automated "firewall" audits that preemptively block reporting during data breaches to maintain organizational trust.
- **Spearheaded Postmortem Root-Cause Analysis (RCA)** for critical data integrity failures, synthesizing business impact reports and collaborating with cross-functional stakeholders to deploy long-term preventative frameworks.
- **Resolved high-complexity, cross-functional BI challenges** by performing advanced statistical analysis to decode intricate data relationships and anomalies, directly influencing strategic business decisions.

**Ameritas Insurance** Cincinnati, Ohio
**Data Governance Intern** September 2021 â€“ April 2022

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program**, establishing the foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, including a centralized Data Dictionary, automated Data Quality rules, and end-to-end Data Lineage to ensure regulatory compliance and transparency.

**Accenture** Pune, India
**Test Engineering Analyst** September 2017 â€“ December 2020

- **Orchestrated end-to-end Quality Assurance** for mission-critical client applications at **Allianz Insurance**, leveraging **Agile SDLC methodologies** to ensure rapid, high-quality feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for both regression testing and the successful rollout of complex new functionalities.
- **Driven operational excellence by automating core regression test suites**, significantly increasing resource productivity and reducing manual testing overhead through scalable automation frameworks.